Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., VII & ECE A
Course Code & Title: EC8791 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar

Innovative Practice Description

- **Unit / Topic: III / Software Performance Optimization**
- **Course Outcome:** Students are able to optimize the performance of the software
- **Topic Learning Outcome:** Explain the concept of loop optimization, cache optimization, performance optimization strategies
- **Activity Chosen:** Flipped Class Room
- **Justification:** It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
  Specific topic was given to the students learn on their own. Resources like reference book, PPT and videos were already post in canvas. This flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related to Code motion, Induction Variable elimination, Strength reduction and cache behavior. Students are separate into groups where students are given a task to perform. Get the class back together to share the individual group's work with everyone.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO2
CO5	2	2	2	2	1	1	-	3

PO / PSO mapped:

Innovative practice	PO9	PO10	PSO2
	1	1	3
Justification for correlation	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	Students delivered an oral presentation about software optimization during this activity	Students will be able to develop software that satisfies end-users. So it is mapped with level 2.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    Students feel that they have improved self-learning. They learn how to communicate with team members and work together.
  - ***Benefit of the practice:***
    Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 70% of students answered correct
  - ***Challenges faced in implementation:***
    The main challenge faced is that few students not exposed to flipped class room. Students struggle with self-discipline and may turn up to class without having absorbed the lesson. To get all students involved, I started to praise the students, took turns for equal participation and joint decision-making.

References:

- Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.
- Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html.

Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)
Degree, Semester & Branch: B.E., V & ECE B
Course Code & Title: EC 8791 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar

Innovative Practice Description

- **Unit / Topic: Unit IV/ Fault Tolerance Technique**
- **Course Outcome:** The students will able to demonstrate the basic concepts of real-time system for embedded system design.
- **Topic Learning Outcome:** Explain fault tolerance techniques, reliability, evaluation, and clock synchronization.
- **Activity Chosen:** Fraternize Group
- **Justification:**

The activity 'Fraternize Group' is used to recall the concepts learnt and enhance the learning through this collaboration. Through this activity the friendly relationship between students can develop. It encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity.

Time Allotted for the Activity: 1 Hour 30 minutes

- **Details of the Implementation:**

The lessons are divided into subcategories. Students are divided into groups of 6 or 7 Students. The Class Strength is divided into 7 Groups with 6 members. The 7 groups are named as 1,2,3,4,5,6,7. Each group allotted with one topic, student learns about his or her topic, preparation materials are posted in CANVAS and presents it to group. Next, Expert group is formed consist of members from each group and discuss their topic with its member 15 Minutes, which has been already discussed with home group. Next, students gather into groups divided by topic. Expert group member presents topics to home group members. It is a cooperative learning method that brings about both individual accountability and achievement of the team goals. Each of these group is given a different topic and allowed to learn about it.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO9	PSO2
CO4	3	2	2	2	1	3

PO / PSO mapped:

Innovative practice	PO9	PSO2
	1	2
Justification for correlation	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	The student will able to design an real time system to satisfy the end-user, So it is mapped at level 2.

Images / Screenshot of the practice:

	
	

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    Students feel that it was interesting. They learn how to communicate with team members and work together.
  - ***Challenges faced in implementation:***
    The main challenge faced is group conflict. Few Students are not ready to work together.

- ❖ ***Benefit of the practice:***

  Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 70% of students answered correct

References:

- https://www.workplacefairness.org/blog/2007/02/06/the-underrated-importance-of-fraternizing/

Signature of Faculty Member **HOD**

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Department of Electronics and Communication Engineering
Academic Year: 2021-2022 (Odd Semester)

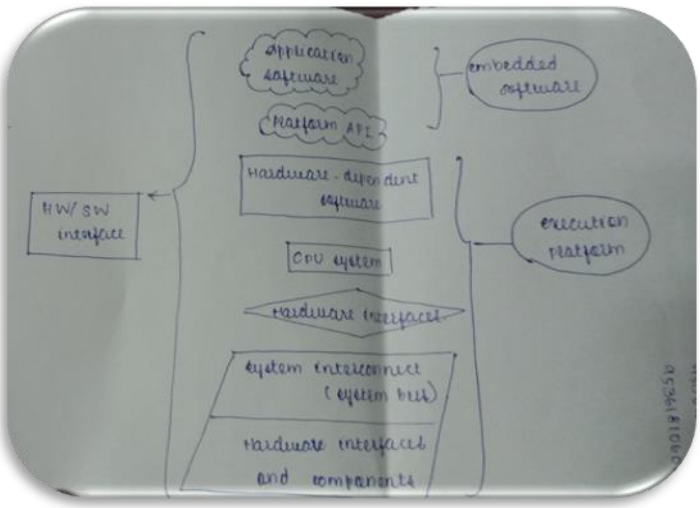
Innovative Practices Description for Unit-I

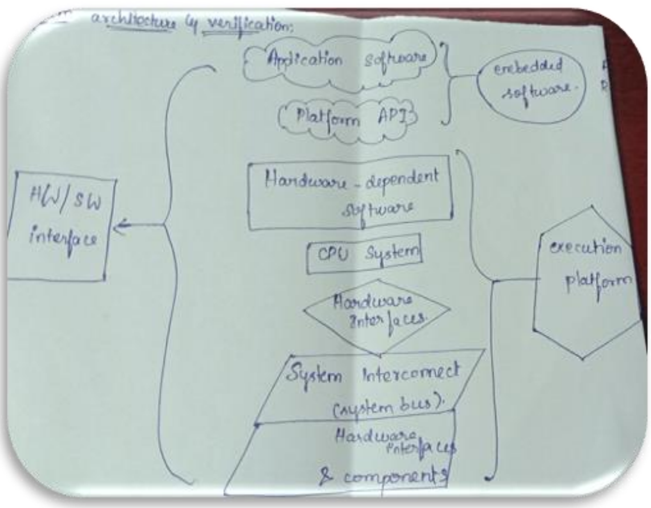
Degree, Semester& Branch: VII Semester B.E. ECE A
Course Code & Title: EC8791 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
Design Example: Model Train Controller - Design methodologies	Story Board	24.08.2021 (10 Minutes)
Specifications-System Analysis and Architecture Design	Mind Map	27.08.221 (10 Minutes)
Platform-Level Performance Analysis	Open Book Test	01.09.2021 (15 Minutes)

Story Board for Design Example: Model Train Controller - Design methodologies on (10 Minutes)


Mind Map for Specifications-System Analysis and Architecture Design on (10 Minutes)





Open Book Test for Platform-Level Performance Analysis
(15 Minutes)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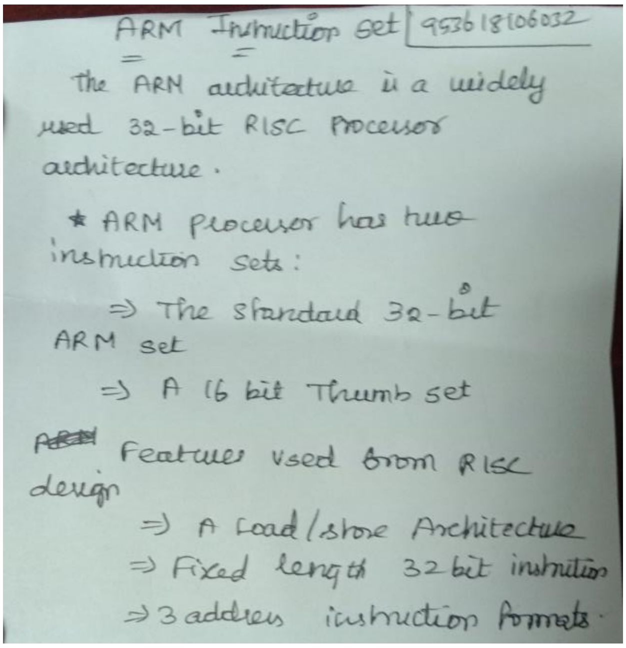
Department of Electronics and Communication Engineering
Academic Year: 2021-2022 (Odd Semester)
<u>Innovative Practices Description for Unit-II</u>

Degree, Semester& Branch: VII Semester B.E. ECE A
Course Code & Title: EC8791 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
Instruction Set of ARM LPC2148	Concept Tests	06.09.2021 (10 Minutes)
Pulse Width Modulation Unit	Visual Quiz	14.09.2021 (10 Minutes)
ARM Cortex M3 and MCU	In-Class Teams	20.09.2021 (15 Minutes)

Concept Tests for Instruction Set of ARM LPC2148


ARM Instruction set 953618106032
The ARM architecture is a widely used 32-bit RISC Processor architecture.
* ARM processor has two instruction sets:
=> The standard 32-bit ARM set
=> A 16 bit Thumb set
Features used from RISC design
=> A Load/store Architecture
=> Fixed length 32 bit instruction
=> 3 address instruction formats.

Visual Quiz for Pulse Width Modulation Unit (10 Minutes)

In-Class Teams for ARM Cortex M3 and MCU (15 Minutes)

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Department of Electronics and Communication Engineering
Academic Year: 2021-2022 (Odd Semester)

Innovative Practices Description for Unit-III

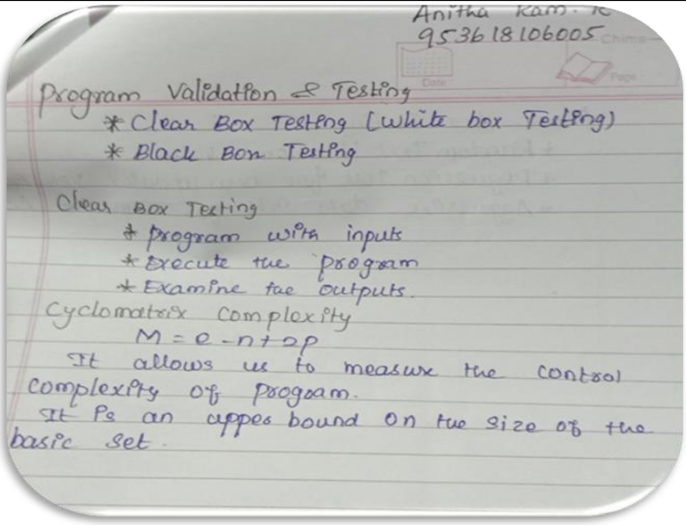
Degree, Semester& Branch: VII Semester B.E. ECE A
Course Code & Title: EC8791 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
Models of Programs	Role-play	23.09.2021 (10 Minutes)
Program Validation and Testing	Exit Slip	05.10.2021 (10 Minutes)

Role-play for Models of Programs

Software Performance Optimization

Exit Slip for Program Validation and Testing



Anitha Kam. K
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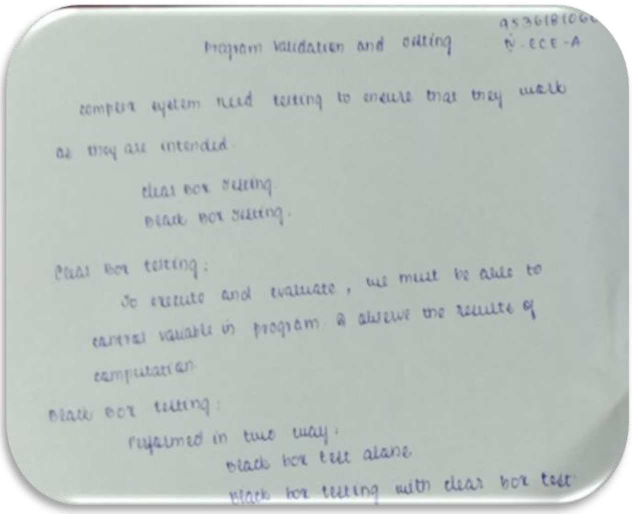
Program Validation & Testing
* Clear Box Testing (white box Testing)
* Black Box Testing

Clear Box Testing
* program with inputs
* Execute the program
* Examine the outputs.

Cyclomatrix complexity
$M = e - n + 2p$
It allows us to measure the control complexity of program.
It is an upper bound on the size of the basic set.



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IV-ECE-A

Program validation and testing

complex system need testing to ensure that they work as they are intended.

clear box testing
black box testing.

Clear box testing:
To execute and evaluate, we must be able to control variable in program & observe the results of computation

Black box testing:
performed in two way:
black box test alone
black box testing with clear box test

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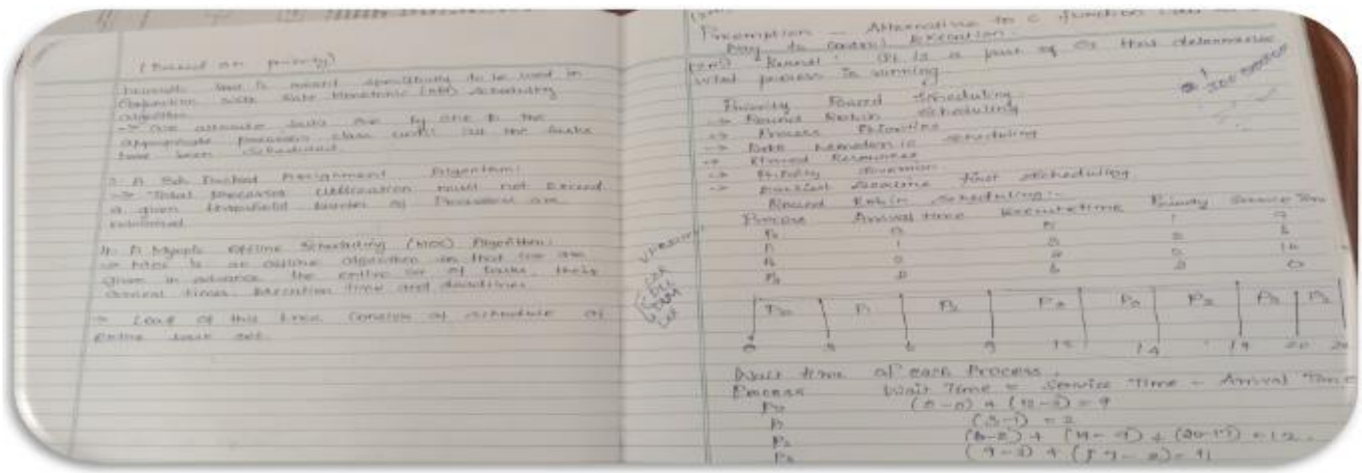
Department of Electronics and Communication Engineering
Academic Year: 2021-2022 (Odd Semester)
<u>Innovative Practices Description for Unit-IV</u>

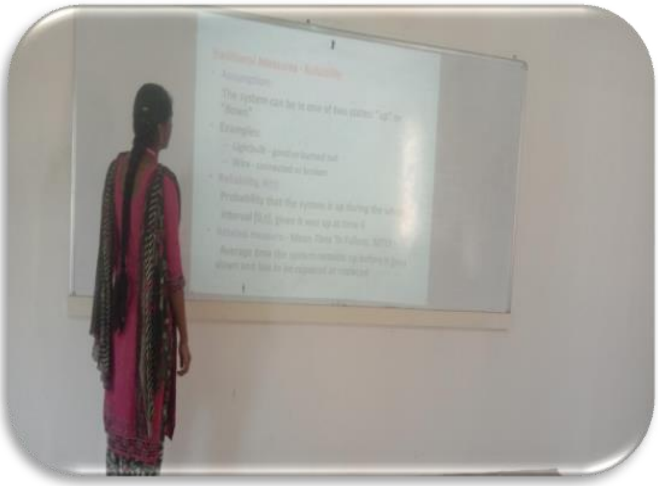
Degree, Semester& Branch: VII Semester B.E. ECE A
Course Code & Title: EC8791 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic	Name of the Innovative Practice	Date & Duration
Task Assignment	Note Check	11.10.2021 (10 Minutes)
Reliability	Story Board	20.10.2021 (10 Minutes)
Clock Synchronization	Open Book Test	25.10.2021 (15 Minutes)

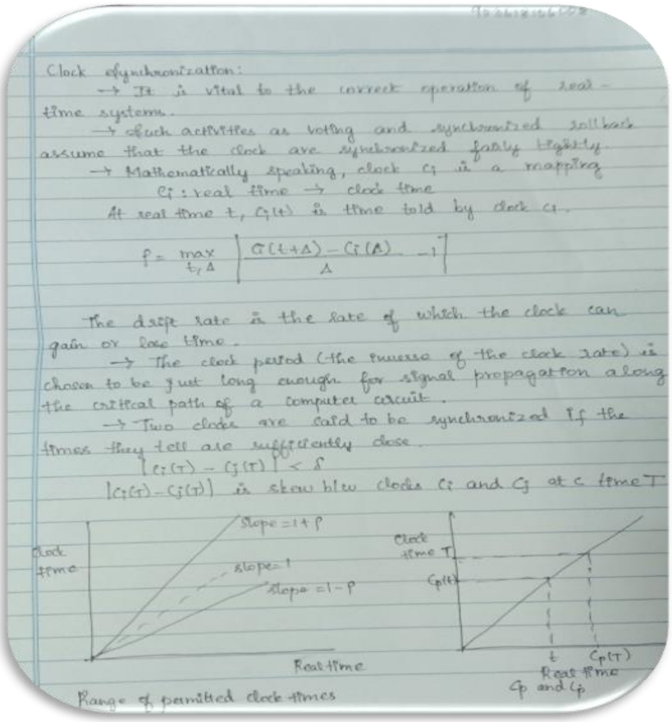
Note Check for Task Assignment



Story Board for Reliability (10 Minutes)



Open Book Test for Clock Synchronization (15 Minutes)



Clock synchronization:

→ It is vital to the correct operation of real-time systems.

→ Such activities as voting and synchronized rollback assume that the clock are synchronized fairly tightly.

→ Mathematically speaking, clock C_i is a mapping

C_i : real time → clock time

At real time t, $C_i(t)$ is time told by clock C_i.

$$\rho = \max_{t,\Delta} \left| \frac{C_i(t+\Delta) - C_i(\Delta)}{\Delta} - 1 \right|$$

The drift rate is the rate of which the clock can gain or lose time.

→ The clock period (the inverse of the clock rate) is chosen to be just long enough for signal propagation along the critical path of a computer circuit.

→ Two clocks are said to be synchronized if the times they tell are sufficiently close.

$$|C_i(T) - C_j(T)| < \delta$$

$|C_i(T) - C_j(T)|$ is skew b/w clocks C_i and C_j at c time T

Clock time

Slope = 1 + ρ

Slope = 1

Slope = 1 − ρ

Real time

Range of permitted clock times

Clock time T

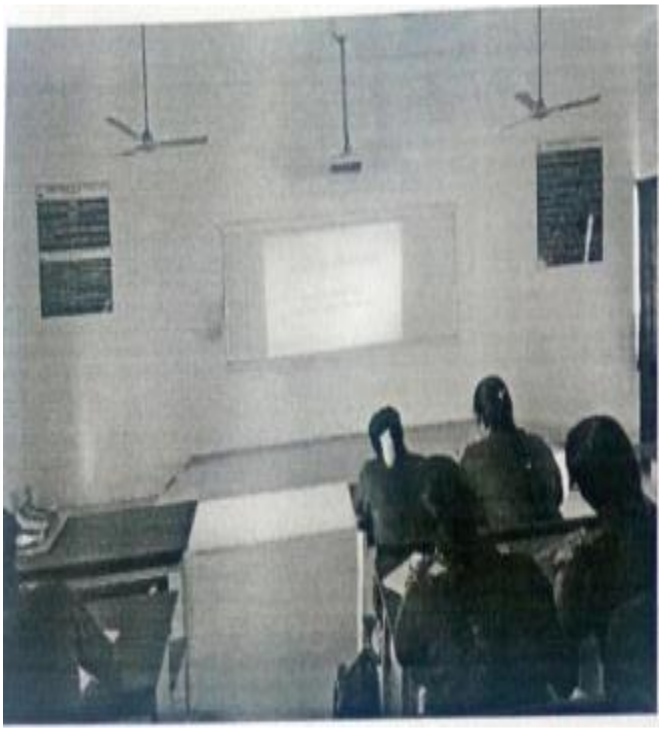
Cp(t)

t

Cp(T)

Real time

Cp and Cp



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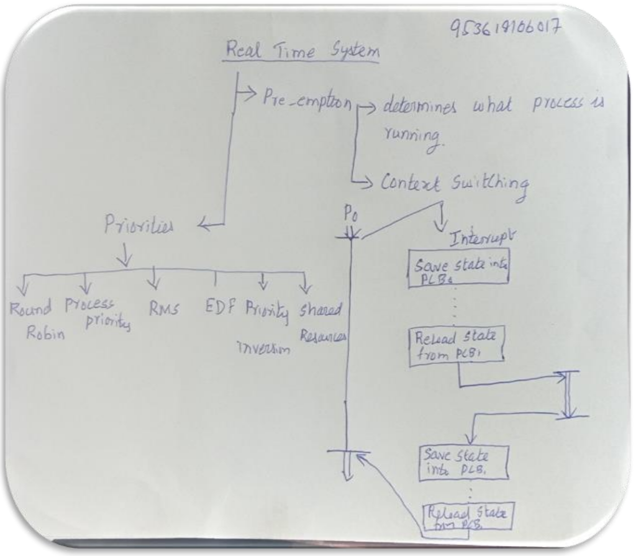
Department of Electronics and Communication Engineering
Academic Year: 2021-2022 (Odd Semester)
<u>Innovative Practices Description for Unit-V</u>

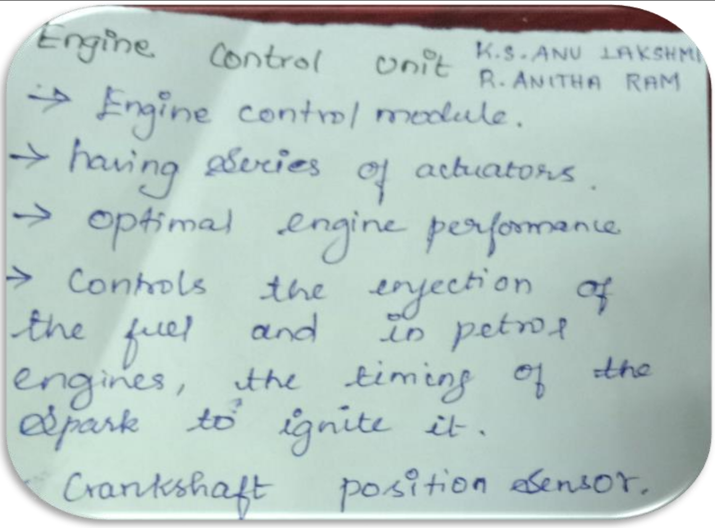
Degree, Semester& Branch: VII Semester B.E. ECE A
Course Code & Title: EC8791 Embedded and Real Time Systems
Name of the Faculty member: Mr.G.Sivakumar

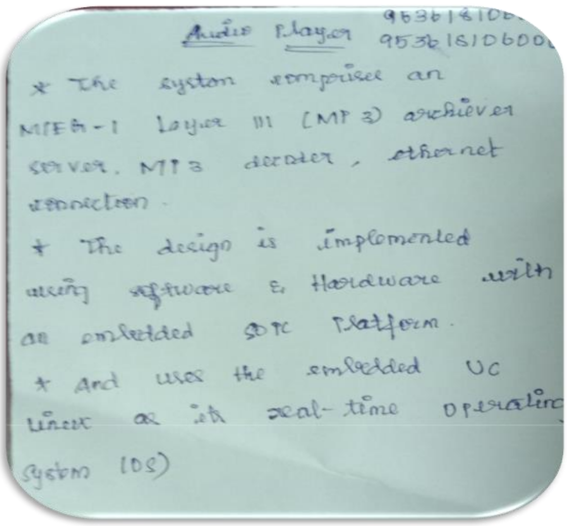
Name of the Topic	Name of the Innovative Practice	Date & Duration
Preemptive Real-Time Operating Systems	Mind Map	01.11.2021 (10 Minutes)
Audio Player, Engine Control Unit, Video Accelerator	Exit Slip	15.11.2021 (10 Minutes)

Mind Map for Preemptive Real-Time Operating Systems



Exit Slip for Audio Player an Engine Control Unit





Signature of Faculty Member **HOD**